CYBERSENTINEL COPILOT - PRESENTATION ARCHITECTURE & DATA FLOW

End-to-End Investigation Architecture: Clear Multi-Agent & Tool Mapping

Frontend UI -> FastAPI Gateway -> 7-Agent LangGraph Pipeline (Aligned to Underlying Engines) -> Verified SOC Output

LAYER 1 : ANALYST INPUT & BACKEND API GATEWAY

How user triggers an investigation and how FastAPI enforces input contracts

1. Security Analyst

SOC Tier-1/2 Analyst
Investigates alerts, scans,
and abnormal network flows

Enters Query

2. Analyst Input Options

* Record ID (e.g. 2039912)
* Free-Text Query (e.g. 'SYN flood')
* Metadata Filters (Protocol/Time)
* Raw Telemetry JSON (78 feats)

UI Trigger

3. SOC Copilot Dashboard

Web UI on http://localhost:8000/
Interactive form, pipeline status,
and live audit cards

HTTP POST

4. FastAPI Microservice

POST /investigate/flow
Pydantic Contract: FlowInvestigationRequest
Input schema & range bounds enforcement

START Orchestration

LAYER 2 : 7-STAGE SEQUENTIAL LANGGRAPH MULTI-AGENT ORCHESTRATION

Sequential deterministic state flow: Every agent executes in strict order with clean state handoffs

Agent 1: Guardrail

* Input Sanitization
* Prompt Injection Filter
* XML Context Isolation
* Relevance Check
* Blocks Auto-Containment

Agent 2: Triage & RAG

* Exact Record Lookup
* 404 Validation Guard
* FTS5 BM25 Search
* Synapt Semantic Search
* RRF Hybrid Ranking

Agent 3: ML Analysis

* HistGradientBoosting
  (9 Attack Families)
* Intrusion Risk (1-P_benign)
* IsolationForest Anomaly
* Top-5 Feature Deviations

Agent 4: Knowledge

* Read-Only incident-agent
* Synapt Knowledge Graph
* trident_query
* trident_get_procedures
* MITRE ATT&CK Mapping

Agent 5: Prioritization

* Deterministic Triage
* Policy Decision Matrix
* Maps Risk to P1-P4
* Evaluates Anomaly Severity
* Discloses Missing Context

Agent 6: Response

* NIST SP 800-61 Playbook
* Step-by-Step Checklist
* Containment Guidance
* Human-in-Loop Safeguards
* Source Provenance

Agent 7: Judge Review

* Read-Only judge-agent
* Hard Invariant Safety Check
* External LLM-as-a-Judge
* Scores 4 Dimensions:
  Grounding/Safety/NIST/Rel

Safety Gate | Exact & BM25 | ML Inference | MCP Tools | Triage Rules | Playbooks | LLM Audit

LAYER 3 : SUPPORTING DATA STORES, ML MODELS, SYNAPT CONTEXT SUBSTRATE & LLM ENGINES

Every agent maps straight down to its exact underlying tool or service with zero line-crossing

Guardrail Engine

src/cyber_assistant/
guardrails.py

* Parameter Range Bounds
* Regex Injection Scanners
* Non-Destructive Invariant
* Domain Term Verification

Local SQLite & FTS5

data/cicids2017.sqlite3

* 2,313,810 Flow Records
* Read-Only Queries
* SQLite FTS5 Index
* 291 Seeded Summaries
* BM25 Keyword Search

ML Model Engines

models/
attack_classifier.joblib
anomaly_detector.joblib

* HistGradientBoosting
  (65 Flow Features)
* IsolationForest (Benign)
* Feature Baseline Profiler

Synapt Context MCP

Provider: assist_cyber
(https://beta.synapt.ai)

* trident_query
* trident_search
* trident_get_procedures
* MITRE ATT&CK KG
* trident_feedback

SOC Triage Matrix

src/cyber_assistant/
policy.py

* P1: Critical Volumetric
* P2: High Threat Scans
* P3: Medium Anomalies
* P4: Low / Benign Flows
* Missing Context Flags

NIST SP 800-61 SOP

SOP Playbook Database

* Phase 1: Detection
* Phase 2: Containment
* Phase 3: Eradication
* Phase 4: Recovery
* Provenance Tracing

LLM Judge & DeepEval

NVIDIA NIM / OpenAI
Endpoint (nvapi-...)

* meta/llama-3.2-11b
* ResponseJudge Engine
* DeepEval v4.2.8 Bench
* 40/40 Passing Cases
* 100% Safety Compliance

LAYER 4 : FINAL VERIFIED OUTPUT & SOC DASHBOARD VISUALIZATION

Complete JSON response rendered into interactive analyst cards and continuous feedback loop

Verified Investigation State Output

FlowInvestigationResponse

* Formatted JSON Schema
* Deterministic Telemetry
* Supervised Probabilities
* Anomaly Deviation Ratios
* MITRE ATT&CK Badges
* NIST Action Playbook
* Verified Judge Verdict
* Complete Audit Trail

Triage Banner

* Attack Family Tag
* Severity Level
* Operational Tier:
  P1, P2, P3, or P4
* Status: Verified
* Non-Destructive Aid

ML Risk Gauge

* Supervised Class
* Intrusion Risk Score:
  1 - P(benign)
* Multiclass Probabilities
* Anomaly Indicator:
  IsolationForest Score

Feature Deviations

* Top-5 Deviations
* Observed Flow Value
* Benign Baseline Mean
* Ratio Multipliers
* Explainable Telemetry

MITRE & Similar

* MITRE ATT&CK IDs
  (e.g. T1046, T1498)
* Tactical Category
* Top Similar Incidents
  (RRF Fused BM25 +
  Synapt Semantic)

NIST Action Plan

* Step-by-Step Actions
* Safe Containment
* Target Host Audit
* Service Isolation Prep
* Human Approval Gate

Judge Quality Card

* Pass Verdict: TRUE
* Safety Score: 100%
* Grounding: 90%
* NIST Align: 85%
* Relevancy: 90%
* LLM Critique Box